

Stuti Shah	
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PROFESSIONAL EXPERIENCE	
Software Engineering Intern, Volvo Group Greensboro, NC	May 2025 - Aug 2025
- Built distributed web applications using Python, Flask, PostgreSQL, and Kubernetes, leveraging GitOps with Argo CD for reliable, reproducible production deployments and environment consistency. - Designed and deployed automated engineering services, reducing workflow execution from 4 hours to under 6 minutes, generating \$176,980 in annual operational savings. - Developed secure, scalable REST APIs integrating high-volume vehicle test data with analytics pipelines, enabling sub-second querying, traceability, and faster engineering decisions. - Integrated ML models into product features, including Random Forest-based component performance predictor and emissions prediction, improving insights on production dashboards.	
AI Intern, AquaOrange Bangkok, Thailand	Jun 2024 - Aug 2024
- Automated AI-driven test case generation for CRM systems, enhancing defect detection by 20%. - Utilized AI testing tools for frontend and backend validation, reducing regression risk and improving stability. - Integrated automated testing into CI pipelines, accelerating issue detection and enhancing release reliability.	
Solutions Associate, HERE! Wireless USA Dallas, TX	Jun 2023 - Jul 2024
- Conducted comprehensive alpha testing and root-cause analysis for Android and iOS apps, improving parental control feature stability and reducing critical crash reports by 35%. - Collaborated with product managers and engineers to translate customer feedback into actionable engineering requirements, accelerating feature refinement and enhancing user satisfaction across 200+ active users. - Performed structured analysis of pricing, usage, and customer behavior data, generating insights that informed product decisions and optimized feature adoption, reducing churn risk by measurable margins.	
CS Undergraduate Teaching Assistant, Virginia Tech Blacksburg, VA	Aug 2024 - May 2025
Provided mentorship to over 200 students in low-level systems topics, utilizing a structured curriculum that covered processor architecture, memory management, and performance optimization concepts, resulting in a 30% increase in student comprehension and application of these critical subjects as evidenced by improved exam scores.	
EDUCATION	
Virginia Tech, Blacksburg, VA - Master of Engineering, Computer Science	May 2026
Virginia Tech, Blacksburg, VA - Bachelor's, Computer Science	May 2025
PROJECTS	
Location-Aware Task Management App	
Built an MVVM-based Android application using Room, LiveData, and ViewModel, enabling offline-first CRUD workflows and location-aware task execution with predictable, low latency user interactions.	
CS Source JobHub	
Developed secure backend services and REST APIs using Flask and MySQL, supporting role-based access, membership lifecycle management, and scalable workflows with strong data integrity guarantees.	
YOLOv8 Shelf Detection for Planogram Compliance (Items + Empty Spaces)	
Built an end-to-end shelf monitoring pipeline using YOLOv8 to detect products and empty shelf spaces from aisle images, curating/training on public datasets to support planogram compliance and restocking insights.	
SKILLS	
Programming Languages : Python, Java, Kotlin, C	
Backend & Systems : Flask, REST APIs, Distributed Systems, Authentication & Authorization, Role-Based Access Control (RBAC), Data Validation, Microservices	
Databases & Data Management : PostgreSQL, MySQL, Room (Android ORM), Relational Schema Design, Query Optimization, Data Integrity, Transaction Management	
Mobile & Frontend : Android SDK, Jetpack Components (LiveData, ViewModel), MVVM Architecture, Geolocation APIs, Offline-First Design, CRUD Workflows	
Cloud & DevOps : Docker, Kubernetes, GitOps (Argo CD), CI/CD Pipelines, Linux, Environment Reproducibility	
Data & Machine Learning : Machine Learning Fundamentals, Random Forests, Feature Engineering, Model Integration, Power BI, Data Visualization, Predictive Analytics	
Software Engineering Practices : Data Structures & Algorithms, Object-Oriented Programming (OOP), SDLC, Unit Testing, Logging & Monitoring, Agile/Scrum, Code Review	